

UMEA, Sweden

WMO: 022860

Lat: 63.792N

Lon: 20.283E

Elev: 14

StdP: 101.16

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.8	-20.8	-26.9	0.3	-23.5	-23.7	0.4	-20.6	11.4	-0.3	10.3	-0.4	1.3	330	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	24.7	17.5	22.9	16.5	21.2	15.7	18.9	22.6	17.7	21.3	16.6	19.9	3.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	12.4	20.7	16.2	11.5	19.4	15.1	10.7	18.1	53.7	22.5	50.0	21.5	46.6	19.8	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.3	7.3	-27.3	27.2	3.8	2.4	-30.0	29.0	-32.3	30.4	-34.4	31.8	-37.2	33.5		
			-27.5	20.1	3.8	1.4	-30.2	21.1	-32.4	21.9	-34.6	22.7	-37.3	23.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-6.9	-7.3	-3.5	1.8	7.2	12.3	15.6	14.1	9.6	3.7	-1.1	-4.6	(6)
	DBStd	9.08	6.31	6.71	5.06	3.26	3.61	2.89	2.64	2.90	3.19	3.80	4.70	5.90	(7)
	HDD10.0	2823	526	484	419	247	103	11	0	3	46	197	333	454	(8)
	HDD18.3	5443	784	718	678	497	347	181	92	134	263	454	583	712	(9)
	CDD10.0	433	0	0	0	0	15	80	174	130	33	1	0	0	(10)
	CDD18.3	11	0	0	0	0	0	1	7	3	0	0	0	0	(11)
	CDH23.3	119	0	0	0	0	2	14	80	23	0	0	0	0	(12)
	CDH26.7	13	0	0	0	0	0	1	11	1	0	0	0	0	(13)
(14) Wind	WSAvg	3.1	3.2	3.2	3.3	3.1	3.3	3.3	3.0	2.8	3.0	3.1	3.2	3.1	(14)
(15) Precipitation	PrecAvg	664	56	39	38	34	40	50	76	84	60	64	67	56	(15)
	PrecMax	1032	114	85	77	96	72	98	186	155	129	131	149	132	(16)
	PrecMin	446	23	2	10	7	4	17	8	17	15	22	19	10	(17)
	PrecStd	128	23	24	17	22	17	19	45	40	34	28	37	29	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	6.6	9.0	15.6	22.8	25.1	28.2	25.8	19.9	13.4	9.3	6.2	(19)
		MCWB	3.5	3.6	4.7	8.9	13.8	15.6	18.8	18.6	15.0	10.1	7.5	4.7	(20)
	2%	DB	3.5	4.4	6.9	12.0	19.4	22.2	25.4	23.7	18.0	11.8	7.6	4.9	(21)
		MCWB	2.1	2.2	3.2	6.6	12.4	14.7	18.1	17.4	13.7	9.7	6.7	3.9	(22)
	5%	DB	2.2	2.6	5.0	9.9	16.7	20.3	23.7	21.9	16.8	10.7	6.2	3.9	(23)
		MCWB	1.3	1.0	2.0	5.5	10.5	13.5	17.2	16.8	13.3	9.3	5.4	3.1	(24)
	10%	DB	1.1	1.2	3.1	7.9	14.4	18.9	22.0	20.1	15.3	9.7	5.2	2.9	(25)
		MCWB	0.5	0.4	0.8	4.1	8.8	12.7	16.3	15.7	12.5	8.3	4.4	2.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.9	5.1	9.3	14.9	17.3	20.7	20.4	16.2	11.7	7.9	5.0	(27)
		MCDB	4.9	6.3	8.6	15.0	21.3	22.5	25.1	23.6	18.2	12.5	8.6	5.5	(28)
	2%	WB	2.3	2.5	3.5	7.2	12.8	15.7	19.3	18.7	15.0	10.6	6.8	4.0	(29)
		MCDB	3.2	4.0	6.3	11.0	18.5	20.2	23.3	22.1	16.9	11.3	7.4	4.5	(30)
	5%	WB	1.4	1.2	2.2	5.7	11.0	14.6	18.3	17.5	14.0	9.5	5.6	3.0	(31)
		MCDB	2.0	2.0	4.3	9.2	15.8	18.8	22.2	20.5	15.9	10.3	6.1	3.5	(32)
	10%	WB	0.5	0.4	1.2	4.3	9.6	13.7	17.2	16.3	13.1	8.3	4.5	2.0	(33)
		MCDB	1.0	1.0	2.6	7.2	13.5	17.6	21.0	19.0	14.9	9.3	5.1	2.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.5	8.9	8.9	10.1	9.9	9.6	9.7	9.2	7.1	5.6	6.7	(35)
		MCDBR	6.9	8.5	9.5	13.0	14.2	13.1	13.1	12.0	10.9	8.6	5.6	5.9	(36)
	5% WB	MCWBR	6.3	7.1	6.7	8.4	8.0	6.8	7.0	7.0	7.4	6.6	5.2	5.4	(37)
		MCWBR	6.6	6.8	8.5	12.0	13.4	11.1	11.1	10.4	9.0	6.9	5.2	5.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.224	0.286	0.311	0.348	0.350	0.349	0.367	0.355	0.326	0.310	0.360	0.350	(40)
		taud	2.025	2.175	2.251	2.319	2.450	2.489	2.462	2.490	2.551	2.403	2.648	2.608	(41)
	Ebn at Noon		363	628	777	828	864	871	842	819	776	614	340	187	(42)
		Edh at Noon	28	61	85	100	96	95	95	85	65	50	25	16	(43)
(44) All-Sky Solar Radiation	RadAvg		0.13	0.71	2.21	3.82	5.08	5.73	5.25	3.97	2.34	0.90	0.21	0.06	(44)
	RadStd		0.01	0.14	0.25	0.35	0.50	0.44	0.49	0.35	0.25	0.17	0.04	0.01	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page